

Learning Cuda Notes

-- global -- function

- means function runs on gpu
- called a kernel
- launched from cpu
- runs many of them in parallel



threadIdx.x

- ID of thread

launching kernel

Global function << # of blocks, # of threads per block >>();

Cuda device Synchronize

- When launching kernel
 - GPU will run asynchronously
 - Out of sync with CPU
- CPU has synchronize and waits for gpu to finish
- Don't overuse, kernels will start in order if chained

Parallelizing Work to Run on GPUs *

Warp Divergence

- When threads in the same warp disagree on conditions
 - Has to run both paths one at a time now
 - Messes with parallelism
- In general
 - Avoid branching a lot in kernels
 - Try to make all threads agree
 - Edge Case branching is fine
 - Don't make branch condition on threadIdx such that it splits the warp
 - Branching on blockIdx is fine since it affects the all warps in a block